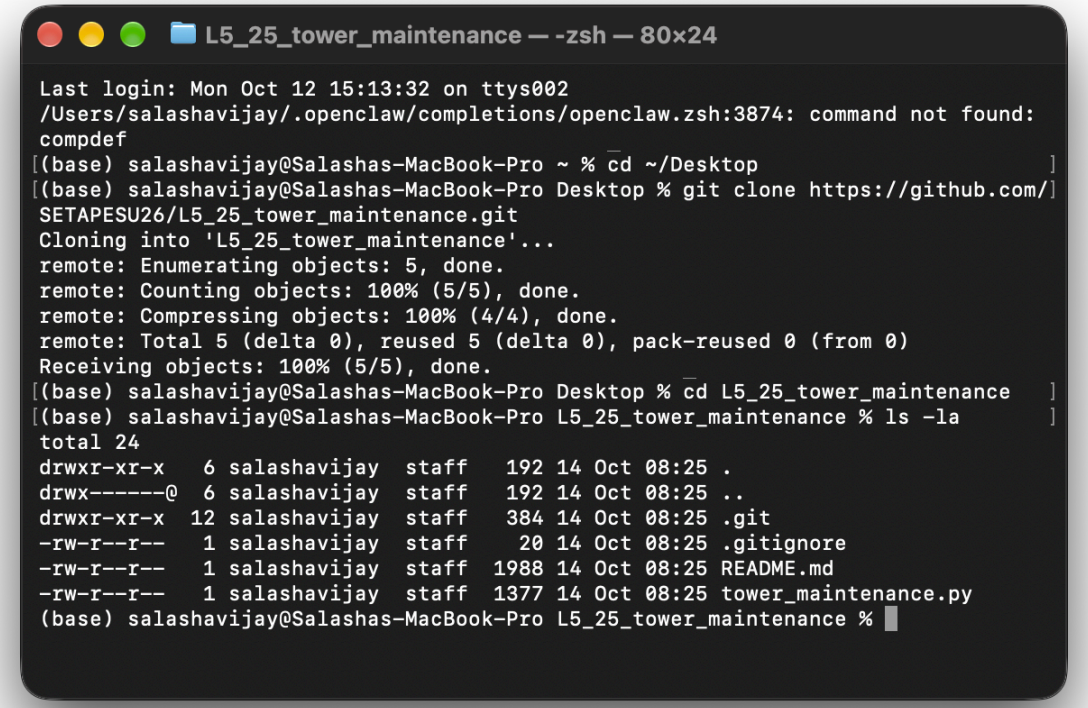
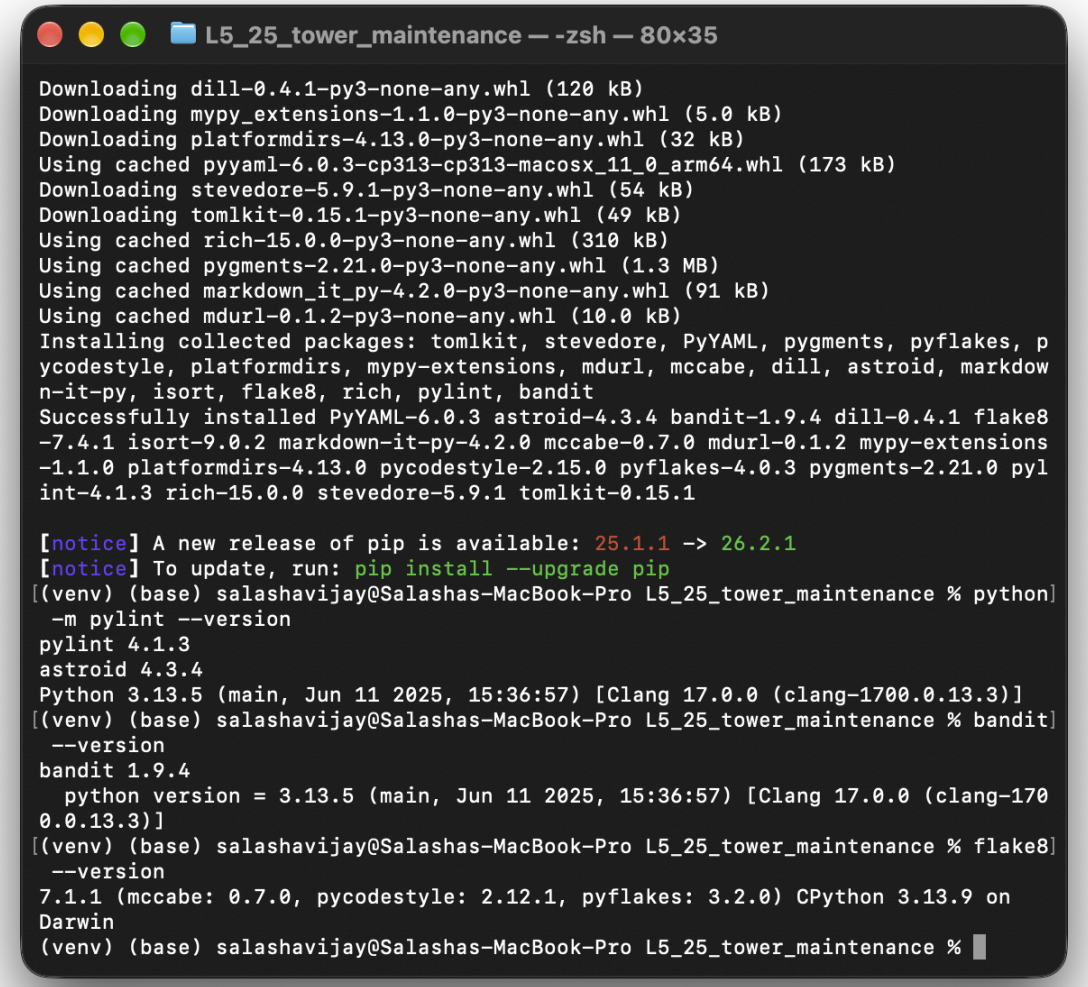


Repository setup



environment setup

```
astroid 4.3.4
Python 3.13.5 (main, Jun 11 2025, 15:36:57) [Clang 17.0.0 (clang-1700.0.13.3)]
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % bandit --version
bandit 1.9.4
  python version = 3.13.5 (main, Jun 11 2025, 15:36:57) [Clang 17.0.0 (clang-1700.0.13.3)]
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % flake8 --version
7.1.1 (mccabe: 0.7.0, pycodestyle: 2.12.1, pyflakes: 3.2.0) CPython 3.13.9 on
Darwin
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat tower_maintenance.py
import pickle
import sqlite3

conn = sqlite3.connect(":memory:")
conn.execute("CREATE TABLE towers (id INTEGER PRIMARY KEY, site TEXT, last_service INTEGER, faults INTEGER)")
NOC_PASSWORD = "tower-noc-77"
service_count = 0


def addTower(site, last_service, faults=0):
    conn.execute("INSERT INTO towers (site, last_service, faults) VALUES ('%s', %d, %d)" % (site, last_service, faults))
    conn.commit()


def serviceTower(tower_id, year):
    global service_count
    try:
        conn.execute("UPDATE towers SET last_service = ?, faults = 0 WHERE id = ?", (year, tower_id))
        service_count += 1
    except Exception:
        print("service failed")



def overdue(year):
    rows = conn.execute("SELECT site, last_service FROM towers").fetchall()
    result = []
    for site, last in rows:
        if year - last >= 2:
            result.append(site)
    return result


def riskLevel(tower_id):
    row = conn.execute("SELECT faults FROM towers WHERE id = ?", (tower_id,)).fetchone()
    if row[0] > 3:
        return "high"
    else:
        return "normal"


def snapshot():
    return pickle.loads(pickle.dumps(conn.execute("SELECT * FROM towers").fetchall()))


def main():
    addTower("Ridge-A", 2021, 5)
    addTower("Coast-B", 2024)
    serviceTower(2, 2025)
    total=len(snapshot())
    print(overdue(2025), riskLevel(1), total, service_count)


main()
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % python tower_maintenance.py
['Ridge-A'] high 2 1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
```

Program before

Before reports

```
main()
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % python tower_maintenance.py
['Ridge-A'] high 2 1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % pylint tower_maintenance.py > pylint_before.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % bandit -r tower_maintenance.py > bandit_before.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % flake8 tower_maintenance.py > flake8_before.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % ls -l *_before.txt
-rw-r--r--  1 salashavijay  staff  2726 14 Oct 08:31 bandit_before.txt
-rw-r--r--  1 salashavijay  staff   467 14 Oct 08:31 flake8_before.txt
-rw-r--r--  1 salashavijay  staff  1930 14 Oct 08:31 pylint_before.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat pylint_before.txt
************* Module tower_maintenance
tower_maintenance.py:5:0: C0301: Line too long (109/100) (line-too-long)
tower_maintenance.py:11:0: C0301: Line too long (120/100) (line-too-long)
tower_maintenance.py:18:0: C0301: Line too long (101/100) (line-too-long)
tower_maintenance.py:1:0: C0114: Missing module docstring (missing-module-docstring)
tower_maintenance.py:7:0: C0103: Constant name "service_count" doesn't conform to UPPER_CASE naming style (invalid-name)
tower_maintenance.py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
tower_maintenance.py:10:0: C0103: Function name "addTower" doesn't conform to snake_case naming style (invalid-name)
tower_maintenance.py:11:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
tower_maintenance.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
tower_maintenance.py:15:0: C0103: Function name "serviceTower" doesn't conform to snake_case naming style (invalid-name)
tower_maintenance.py:16:4: W0603: Using the global statement (global-statement)
tower_maintenance.py:20:11: W0718: Catching too general exception Exception (broad-exception-caught)
tower_maintenance.py:25:0: C0116: Missing function or method docstring (missing-function-docstring)
tower_maintenance.py:34:0: C0116: Missing function or method docstring (missing-function-docstring)
tower_maintenance.py:34:0: C0103: Function name "riskLevel" doesn't conform to snake_case naming style (invalid-name)
tower_maintenance.py:36:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
tower_maintenance.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)
tower_maintenance.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 5.14/10

(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat bandit_before.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
Run started:2026-10-14 03:01:29.224625+00:00

Test results:
>> Issue: [B403:blacklist] Consider possible security implications associated with pickle module.
   Severity: Low   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b403-import-pickle
   Location: ./tower_maintenance.py:1:0
1       import pickle
2       import sqlite3
3

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'tower-noc-77'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: ./tower_maintenance.py:6:15
5       conn.execute("CREATE TABLE towers (id INTEGER PRIMARY KEY, site TEXT, last_service INTEGER, faults INTEGER)")
6       NOC_PASSWORD = "tower-noc-77"
7       service_count = 0

--------------------------------------------------
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: ./tower_maintenance.py:11:17
10      def addTower(site, last_service, faults=0):
11          conn.execute("INSERT INTO towers (site, last_service, faults) VALUES ('%s', %d, %d)" % (site, last_service, faults))
12          conn.commit()

--------------------------------------------------
>> Issue: [B301:blacklist] Pickle and modules that wrap it can be unsafe when used to deserialize untrusted data, possible security issue.
   Severity: Medium   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b301-pickle
   Location: ./tower_maintenance.py:43:11
42      def snapshot():
43          return pickle.loads(pickle.dumps(conn.execute("SELECT * FROM towers").fetchall()))
44

--------------------------------------------------

Code scanned:
        Total lines of code: 38
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 2
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 2
                High: 2
Files skipped (0):
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat flake8_before.txt
tower_maintenance.py:5:80: E501 line too long (109 > 79 characters)
tower_maintenance.py:11:80: E501 line too long (120 > 79 characters)
tower_maintenance.py:18:80: E501 line too long (101 > 79 characters)
tower_maintenance.py:25:1: E303 too many blank lines (3)
tower_maintenance.py:35:80: E501 line too long (88 > 79 characters)
tower_maintenance.py:43:80: E501 line too long (86 > 79 characters)
tower_maintenance.py:50:10: E225 missing whitespace around operator
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
```

After reports

```
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % python tower_maintenance.py
['Ridge-A'] high 2 1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % pylint tower_maintenance.py
************* Module tower_maintenance
tower_maintenance.py:17:0: C0103: Constant name "service_count" doesn't conform to UPPER_CASE naming style (invalid-name)
tower_maintenance.py:34:4: W0603: Using the global statement (global-statement)

-------------------------------------------------------------------
Your code has been rated at 9.46/10 (previous run: 5.14/10, +4.32)

(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % bandit -r tower_maintenance.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
Run started:2026-10-14 03:11:53.648685+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 76
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % flake8 tower_maintenance.py
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
```

```
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % python tower
_maintenance.py
pylint tower_maintenance.py
bandit -r tower_maintenance.py
flake8 tower_maintenance.py
['Ridge-A'] high 2 1

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
Run started:2026-10-14 03:15:51.095773+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 77
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Git Status and Commit History

```
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % pylint tower_maintenance.py > pylint_after.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % bandit -r tower_maintenance.py > bandit_after.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % flake8 tower_maintenance.py > flake8_after.txt 2>&1
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % ls -l *_before.txt *_after.txt
-rw-r--r--  1 salashavijay  staff   620 14 Oct 08:47 bandit_after.txt
-rw-r--r--  1 salashavijay  staff  2726 14 Oct 08:31 bandit_before.txt
-rw-r--r--  1 salashavijay  staff     0 14 Oct 08:47 flake8_after.txt
-rw-r--r--  1 salashavijay  staff   467 14 Oct 08:31 flake8_before.txt
-rw-r--r--  1 salashavijay  staff   140 14 Oct 08:47 pylint_after.txt
-rw-r--r--  1 salashavijay  staff  1930 14 Oct 08:31 pylint_before.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   tower_maintenance.py

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        bandit_after.txt
        bandit_before.txt
        flake8_after.txt
        flake8_before.txt
        pylint_after.txt
        pylint_before.txt

no changes added to commit (use "git add" and/or "git commit -a")
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % git log --oneline -3
8689fd4 (HEAD -> main, origin/main, origin/HEAD) Add Lab 5 static analyis scenario
```

Final reports

```
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat pylint_after.txt

-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat bandit_after.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
Run started:2026-10-14 03:17:48.524279+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 77
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat flake8_after.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
```

```
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % cat flake8_after.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % git add tower_maintenance.py
git commit -m "Fix security issues and improve code quality"
[main eb2705b] Fix security issues and improve code quality
 Committer: Salasha Vijay <salashavijay@Salashas-MacBook-Pro.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 81 insertions(+), 31 deletions(-)
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % git add flake8_before.txt flake8_after.txt
git commit -m "Add Flake8 before and after reports"
[main 332f61f] Add Flake8 before and after reports
 Committer: Salasha Vijay <salashavijay@Salashas-MacBook-Pro.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

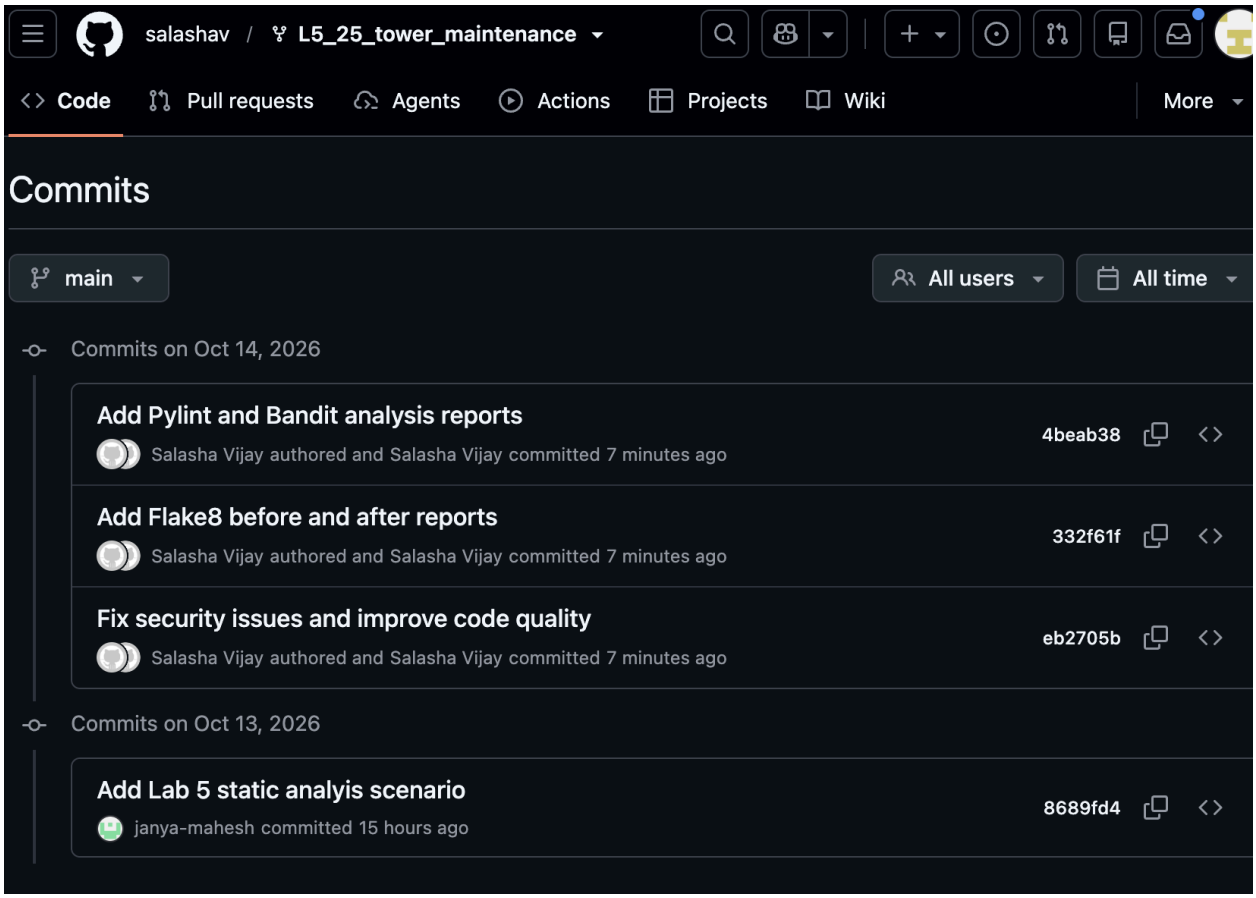
 2 files changed, 7 insertions(+)
 create mode 100644 flake8_after.txt
 create mode 100644 flake8_before.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance % git add pylint_before.txt pylint_after.txt bandit_before.txt bandit_after.txt
git commit -m "Add Pylint and Bandit analysis reports"
[main 4beab38] Add Pylint and Bandit analysis reports
 Committer: Salasha Vijay <salashavijay@Salashas-MacBook-Pro.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 4 files changed, 120 insertions(+)
 create mode 100644 bandit_after.txt
 create mode 100644 bandit_before.txt
 create mode 100644 pylint_after.txt
 create mode 100644 pylint_before.txt
(venv) (base) salashavijay@Salashas-MacBook-Pro L5_25_tower_maintenance %
```


salashav / L5_25_tower_maintenance
Code
Pull requests
Agents
Actions
Projects
Wiki
More
Commits
main
All users
All time
Commits on Oct 14, 2026
Add Pylint and Bandit analysis reports
Salasha Vijay authored and Salasha Vijay committed 7 minutes ago
4beab38
Add Flake8 before and after reports
Salasha Vijay authored and Salasha Vijay committed 7 minutes ago
332f61f
Fix security issues and improve code quality
Salasha Vijay authored and Salasha Vijay committed 7 minutes ago
eb2705b
Commits on Oct 13, 2026
Add Lab 5 static analyis scenario
janya-mahesh committed 15 hours ago
8689fd4